

Enhancing Autonomous Vehicle-Pedestrian Interaction in Shared Spaces: The Impact of Intended Path-Projection

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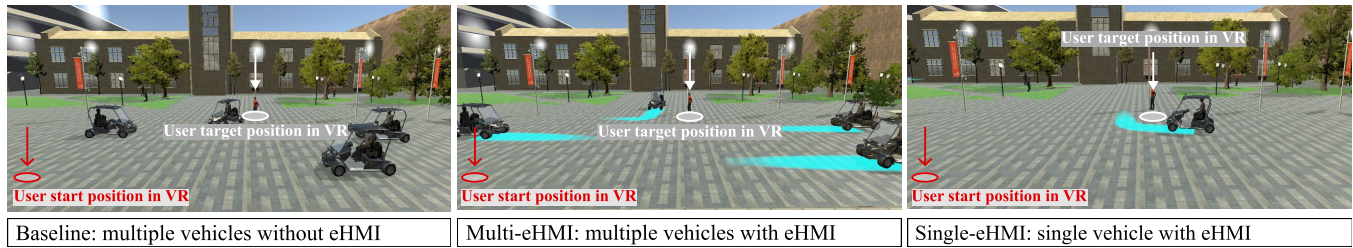


Figure 1: The three experimental conditions in this study: (Left) Baseline – multiple vehicles without eHMI, (Middle) Multi-eHMI – multiple vehicles with eHMI, (Right) Single-eHMI – single vehicle with eHMI.

Abstract

External Human-Machine Interfaces (eHMI) are critical for seamless interactions between autonomous vehicles (AVs) and pedestrians in shared spaces. However, they often struggle to adapt to these environments, where pedestrian movement is fluid and right-of-way is ambiguous. To address these challenges, we propose PavFlow, an eHMI that projects the AV's intended path onto the ground in real time, providing continuous spatial information rather than a binary stop/go signal. Through a VR study (N=18), we evaluated PavFlow's effectiveness under two AV density conditions (single vs. multiple AVs) and a baseline condition without PavFlow. The results showed that PavFlow significantly improved pedestrian perception of safety, trust, and user experience while reducing cognitive workload. This performance remained consistent across both single and multiple AV conditions, despite persistent tensions in priority negotiation. These findings suggest that path projection enhances eHMI transparency by offering richer movement cues, which may better support AV-pedestrian interaction in shared spaces.

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CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Co-navigation, shared space, eHMI, AV-pedestrian interaction, path-projection

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1 Introduction

Shared spaces eliminate traditional vehicle-pedestrian boundaries, necessitating real-time right-of-way negotiation [23, 25, 26, 31, 35], with no party holding absolute priority [32]. This process, known as co-navigation, involves continuous mutual adjustment between road users [48]. In such environments, clear communication becomes essential for safe and efficient interactions. For autonomous vehicles (AVs), External Human-Machine Interfaces (eHMIs) have been proposed to clarify AV intentions and improve AV-pedestrian interactions [9]. However, most existing eHMIs are designed for structured roads [45], typically focusing on one-to-one communication rather than the dynamic, multi-user co-navigation characteristic of shared spaces [31, 45].

In shared spaces, AVs need to dynamically respond to pedestrians in real time [27]. Current AVs rely on internal path adjustment algorithms to achieve this, but the subtlety of their motion cues in low-speed shared spaces can render these adjustments opaque to pedestrians. Furthermore, explicit communication via eHMIs is often reduced to binary responses—either yielding or not yielding when paths may cross [50]—without accommodating more nuanced interactions. Moreover, AV communication typically does not adapt to the continuous nature of pedestrian movements (e.g., altering paths), which limits real-time negotiation in shared environments [45].

To address this gap, we introduce *PaveFlow*, an eHMI that externalises an AV's intended path and responds to pedestrians in real-time. By making an AV's future movements more transparent, *PaveFlow* aims to improve AV-pedestrian interactions in shared spaces. To evaluate the effectiveness and scalability of *PaveFlow*, we conducted a within-subjects Virtual Reality (VR) study (N=18) with three experimental conditions: *PaveFlow* visualisation in two AV density situations (single vs. multiple AVs) and a baseline condition without *PaveFlow*. We analysed participant feedback on perceived safety, trust, workload, and user experience to assess *PaveFlow*'s impact on co-navigation. Results showed that *PaveFlow* significantly enhanced pedestrian confidence and trust, lowered cognitive workload, and performed consistently across single and multi-AV conditions. However, in multi-AV settings, participants reported confusion as different AV motion states appeared simultaneously in *PaveFlow* projections.

The contributions of this paper are twofold:

- Introduces *PaveFlow* as a design concept for a scalable, adaptive eHMI in shared spaces.
- Provides empirical insights into how externalised path information shapes AV-pedestrian interactions, informing future eHMI development.

2 Related Work

2.1 AV-Pedestrian Interaction in Shared Spaces

Prior research on AV-pedestrian interaction has mainly centred on algorithmic approaches to pedestrian handling. Approaches such as game-theoretic models [2], predictive models [40], and proxemics-based methods [3] are theoretically effective but underperform in practice [4]. AV motion planning algorithms often fail to handle pedestrian unpredictability and lack effective communication, leading to overly cautious behaviours like the '*freezing robot problem*' [6].

While evidence shows that eHMIs improve AV-pedestrian interactions [34], few studies focus on eHMIs designed specifically for shared spaces [18, 45]. Existing eHMIs are often developed for structured environments with clear right-of-way rules and one-to-one interactions, making them ill-suited for shared spaces where priority is ambiguous and multiple road users must coordinate movement [31, 45]. This lack of clear communication increases uncertainties and lowers efficiency [22, 39, 44].

Effective co-navigation in shared spaces demands eHMIs that facilitate complex, multi-user negotiations [41, 49]. Existing eHMI designs primarily rely on communication signals such as LED light

bars [7, 17, 28], with 97% of them focusing on conveying the vehicle's yielding intent [9]. While this information is crucial for structured roads and crossing scenarios, it is less effective in shared spaces, where seamless co-navigation depends on pedestrians being able to anticipate AV movements more granularly and adjust their own accordingly. A notable attempt to improve AV-pedestrian communication beyond yielding intent is *Gazing Eyes* [16], which uses anthropomorphic robotic eyes to indicate the vehicle's fine-grained moving direction. However, eye-based communication can be subtle and difficult to notice.

2.2 Projection-Based eHMI

Existing projection-based eHMIs convey AV intentions through ground-projected symbols like stop lines [29, 30, 42], symbols [38], and directional arrows [9, 13]. While factors such as adverse weather [33], bright daylight [30], glare, and uneven surfaces [33] can affect visibility, projection-based eHMIs remain highly adaptable and practical in shared spaces. Nguyen et al. [30] highlighted several advantages: they are visible to multiple pedestrians at once, potentially from different viewing angles, and they utilise the road surface, a familiar medium for conveying traffic information. Building on these strengths, our study explores a dynamic path projection concept that enhances AV-pedestrian interaction.

3 METHODOLOGY

3.1 Design Concept

The design concept, *PaveFlow*, integrates AV's navigation algorithms [21, 47] with a path projection eHMI, as shown in Figure 2. It dynamically projects the AV's intended trajectories onto the ground and adjusts them in real time based on pedestrian movements. *PaveFlow* was designed around three principles derived from a detailed analysis of co-navigation challenges in shared spaces [45].

- *Continuity*: Keep AV intentions constantly updated, helping pedestrians and AVs adjust in real time.
- *Predictability*: Offer clear visual cues that help pedestrians easily understand AV intentions.
- *Flexible Right-of-Way*: Allow AVs and pedestrians to flexibly share the right of way. Instead of AVs always yielding to pedestrians, *PaveFlow* enables a more balanced interaction where pedestrians can decide to yield based on the AV's projected path.

PaveFlow was developed through an iterative process, beginning with low-fidelity sketches and evolving into a high-fidelity 3D Unity prototype where an AV navigation algorithm was developed. Based on this algorithm, the system projects the AV's expected path up to three seconds ahead (see Figure 2, middle), providing pedestrians with clear, anticipatory visual cues.

We selected cyan to convey the AV's intended trajectory, as research supports it as optimal colour for eHMI: it is distinguishable [46], neutral for yielding intent [10], and intuitive for pedestrian crossings [1].

3.2 Experimental Design and Hypotheses

The experiment employed a within-subjects design in which each participant experienced three conditions (see Figure 1):

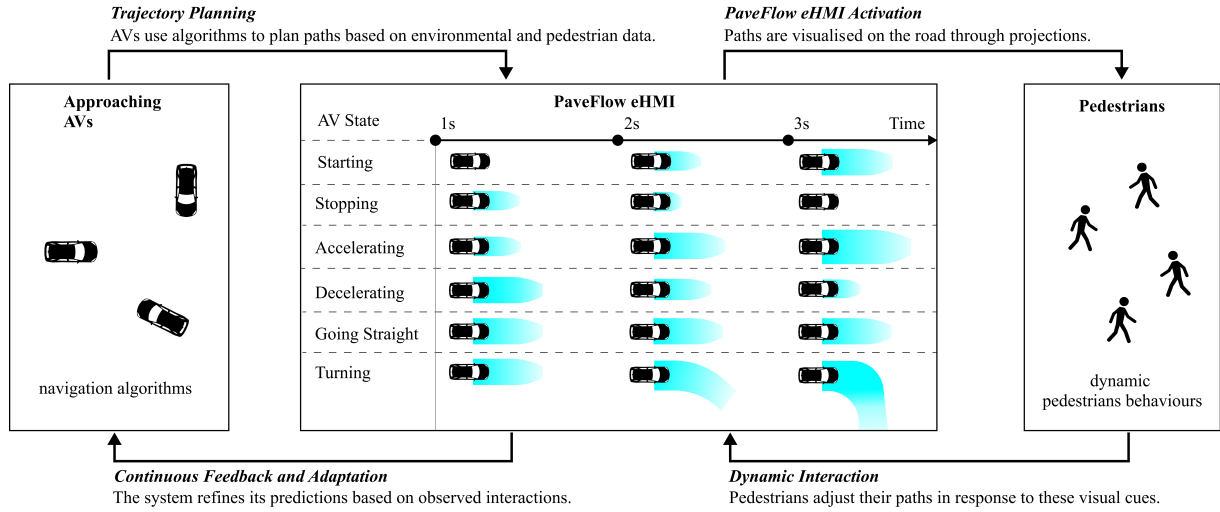


Figure 2: Overview of design concept which integrates AV navigation algorithms with a path projection eHMI to visualise intended trajectories in real time. As pedestrians adjust their movement in response, the concept enables dynamic right-of-way negotiation.

- *Baseline*: Multiple AVs and no eHMI
- *Multi-eHMI*: Multiple AVs with eHMI
- *Single-eHMI*: Single AV with eHMI

The *Baseline* condition serves as a reference for AV-pedestrian interactions in realistic shared spaces with multiple vehicles, ensuring findings remain applicable to real-world scenarios. Using a multi-AV baseline allows us to assess whether eHMI remains effective when pedestrians must interpret multiple AVs simultaneously, rather than in isolated AV-pedestrian encounters. Since AV behaviour was fully controlled in VR, this design choice does not introduce additional variability. The *Multi-eHMI* condition evaluates eHMI effectiveness in these high-density settings. The *Single-eHMI* condition was added to examine how eHMI functions in simpler AV-pedestrian interactions.

Based on the theory of shared mental models [36], which suggests that transparent and predictable communication helps individuals anticipate system behaviour, reduce uncertainty, and build trust, we propose the following hypotheses:

- *H1 (Effectiveness)*: PaveFlow improves AV-pedestrian interactions by enhancing safety and trust, reducing cognitive workload, and improving user experience compared to the baseline condition without eHMI.
- *H2 (Scalability)*: PaveFlow maintains its effectiveness in both single and multiple AV conditions.

3.3 Apparatus

PaveFlow was evaluated through a VR-based experiment. The virtual environment was modelled to replicate an actual shared space within our university campus. The AV used in the experiment was a small, motorised vehicle, typically used in shared spaces [15, 45]. The AV operated at low speeds, up to 10 km/h. Originally, the AV was designed to predict pedestrian movement paths and adjust its behaviour dynamically. However, in the experiment, we simplified

this logic: the AV only adjusted its speed and projected its path based on its distance to pedestrians, without predicting their future movement. This simplification ensured consistent and repeatable interactions, avoiding variability from prediction errors. The study took place in an open floor area with an 11×13-metre movement zone. A Meta Quest 2 VR headset was used to create an immersive simulation.

3.4 Data Collection and Analysis

3.4.1 Quantitative Measures:

- *Perceived Safety*: Assessed using a 5-point Likert Scale adapted from prior research [20].
- *Trust in Automation*: Measured using the Trust in Automation Scale [24].
- *Workload*: Evaluated using the NASA Task Load Index [5].
- *User Experience*: Measured using the short version of the User Experience Questionnaire [37].

Quantitative data were analysed using descriptive statistics to summarise trends and inferential statistics to test for differences across conditions. A global test across all three conditions was not conducted because *Baseline* and *Single-eHMI* differ in both the number of AVs and the presence of eHMI, making direct comparisons less meaningful. Instead, two separate tests were conducted: one-way repeated measures ANOVA for normally distributed data (*Baseline* and *Multi-eHMI*) and the Wilcoxon signed-rank test for non-parametric data (between *Multi-eHMI* and *Single-eHMI*). A significance threshold of $p < 0.05$ was applied for all tests.

3.4.2 Qualitative Measures: Qualitative data were collected through semi-structured interviews conducted post-condition (4-8 minutes each) and post-study (20-30 minutes) to assess participants' perceptions of the PaveFlow eHMI, user experience, and AV behaviour. The interviews focused on participant's overall experience

of interacting with the AVs, the different conditions, and suggestions for design improvement. Thematic analysis was employed to extract key themes and insights, helping to identify recurring patterns and refine the understanding of participant feedback.

3.5 Participants

Eighteen participants (aged 18–34) were recruited using posters, flyers, and social media. The group comprised 16 women, 1 men, and 1 person who did not disclose their gender. Regarding VR and AV familiarity, 8 participants had no prior VR experience and 10 participants were somewhat familiar with AVs (e.g., through popular media).

3.6 Study Procedure

The experimental procedure comprised four phases to ensure participant familiarity and minimise learning effects. In the familiarisation phase, participants adapted to the VR environment and navigation tasks without moving vehicles. During the practice round, they navigated to a stationary target without distractions to learn the VR controls. In the actual experiment, participants completed three randomised conditions, each lasting about three minutes, followed by a questionnaire and brief feedback during a short interview. Finally, the post-experiment session involved a semi-structured interview to discuss experiences across conditions and gather suggestions for improvement. This study was carried out following the ethical approval granted by the University of Sydney (project protocol 2023/HE000434).

4 RESULTS

Figure 3 presents the descriptive statistics along with statistical significance indicators for comparisons across conditions.

4.1 Perceived Safety

A one-way repeated measures ANOVA comparing *Baseline* and *Multi-eHMI* revealed a significant effect of eHMI presence ($F(1,18) = 18.133$, $p < 0.001$), with *Multi-eHMI* leading to higher Perceived Safety. A Wilcoxon signed-rank test comparing *Multi-eHMI* and *Single-eHMI* showed no significant difference ($Z = 1.570$, $p = 0.116$).

Qualitative feedback: In *Baseline*, participants felt ‘surrounded’ (P13) and ‘confused’ (P10, P15) from the lack of cues. Many expressed uncertainty about whether the vehicle would yield ($n=8$) and hesitated to proceed walking due to fear of collision ($n=5$). In *Multi-eHMI*, pedestrians showed improved awareness of AV actions like ‘turning’ and ‘going straight’ ($n=8$) compared to *Baseline*. However, they struggled to accurately identify changes like ‘stopping’ or ‘slowing down’ when multiple eHMIs were present ($n=7$). To improve comprehension and enhance pedestrian responsiveness, participants suggested to incorporate auditory signals for ‘stopping’ ($n=6$) and providing detailed speed information for ‘accelerating/decelerating’ ($n=8$). In *Single-eHMI*, participants actively adjusted their paths and speed ($n=10$) in response to changes in the AV’s movement. Comments such as ‘I believe the vehicle won’t hit me’ (P18) suggest that the PaveFlow’s visual cues provided reassurance and helped pedestrians navigate safely.

4.2 Trust in Automation

A one-way repeated measures ANOVA comparing *Baseline* and *Multi-eHMI* revealed a significant effect of eHMI presence ($F(1,18) = 17.713$, $p < 0.001$), with *Multi-eHMI* leading to higher trust. A Wilcoxon signed-rank test comparing *Multi-eHMI* and *Single-eHMI* showed no significant difference ($Z = 1.659$, $p = 0.097$).

Qualitative feedback: In *Baseline*, changes in AV motion states (e.g., yielding or turning) caused pedestrian anxiety ($n=3$). For instance, pedestrians often misinterpreted ‘yielding to stop’ as an ‘AV malfunction’ (P18), leading to mistrust and tense behaviour. Reactions varied: some waited for all AVs to pass, while others hurried to avoid collisions. In *Multi-eHMI*, pedestrian trust generally increased. However, it is important to note that within this environment, trust decreased during AV state transitions due to challenges in quickly interpreting PaveFlow information ($n=9$). Participants (e.g., P12, P14, P18) suggested that familiarity with PaveFlow could enhance usage confidence. In *Single-eHMI*, pedestrians navigated at a normal pace, instinctively interpreting PaveFlow as the AV’s path ($n=14$).

4.3 Workload

Figure 4 presents descriptive workload ratings across conditions for each NASA-TLX dimension. For overall workload, a one-way repeated measures ANOVA comparing *Baseline* and *Multi-eHMI* revealed a significant effect of eHMI presence ($F(1,18) = 6.253$, $p = 0.023$), with *Multi-eHMI* resulting in lower workload. A Wilcoxon signed-rank test comparing *Multi-eHMI* and *Single-eHMI* also found a significant difference ($Z = -2.628$, $p = 0.009$), with *Single-eHMI* further lowering workload.

Qualitative feedback: In *Baseline*, participants felt ‘overwhelmed’ ($n=6$) and ‘mentally exhausted’ ($n=12$) as they expended excessive effort on ‘avoiding collisions’ and ‘guessing AV intentions’ rather than focusing on navigation. The high mental workload resulted in reduced task efficiency. In contrast, in *Multi-eHMI*, PaveFlow’s ‘intuitive’ ($n=12$) signals allowed pedestrians to easily interpret AV intentions, reducing task complexity and enabling greater focus on navigation, significantly improving mental workload and reducing frustration. In *Single-eHMI*, the reduced number of AVs further eased task demands, making navigation even smoother.

4.4 User Experience

A one-way repeated measures ANOVA comparing *Baseline* and *Multi-eHMI* revealed a significant effect of eHMI presence ($F(1,18) = 15.143$, $p = 0.001$) on pragmatic quality and a significant effect ($F(1,18) = 7.674$, $p = 0.013$) on hedonic quality, with *Multi-eHMI* yielding higher scores in both measures. A Wilcoxon signed-rank test comparing *Multi-eHMI* and *Single-eHMI* showed no significant difference on pragmatic quality ($Z = 1.454$, $p = 0.146$) and hedonic quality ($Z = -0.456$, $p = 0.649$).

Qualitative feedback: In *Baseline*, participants found navigation complicated ($n=15$). In *Multi-eHMI*, PaveFlow was praised for ‘providing valuable information,’ ($n=10$) improving pragmatic quality. The dynamic changes in PaveFlow also sparked curiosity, with participants stating ‘I feel like it reacts because of me,’ (P12) emphasising the role of dynamic feedback in enhancing hedonic quality. In

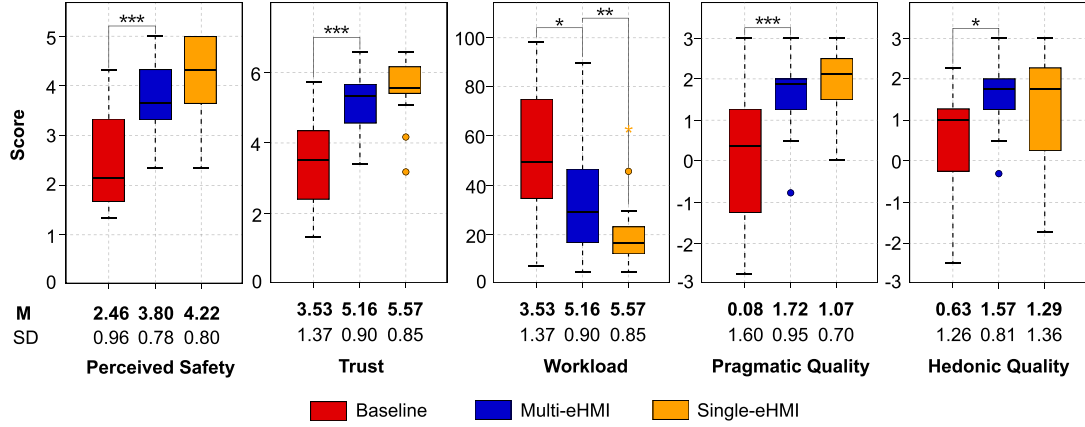


Figure 3: Boxplots show Descriptive Means (M) and Standard Deviations (SD) of Perceived Safety, Trust, Workload, Pragmatic Quality, and Hedonic Quality across conditions. Statistical significance was tested using ANOVA (Baseline vs. Multi-eHMI) and Wilcoxon (Multi-eHMI vs. Single-eHMI). *: $p \leq .05$, **: $p \leq .01$, ***: $p \leq .001$

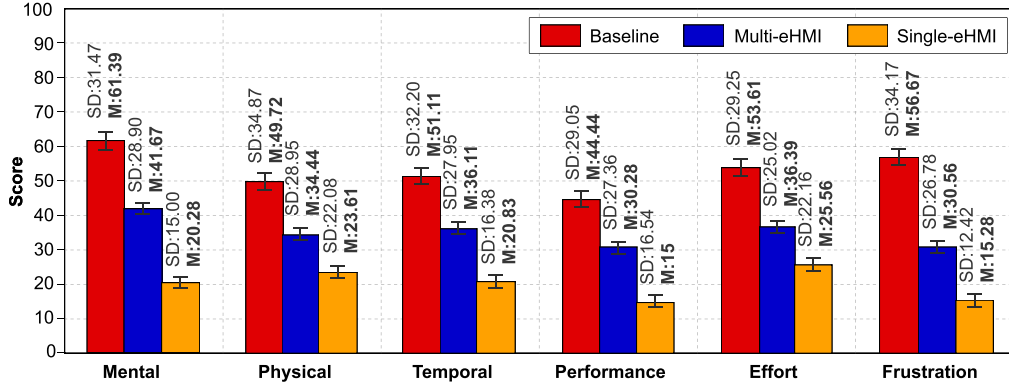


Figure 4: Descriptive statistics of NASA-TLX workload dimensions across the three study conditions.

Single-eHMI, participants described the experience as ‘most supportive’ ($n=5$) and ‘easiest,’ ($n=16$) considering it the most user-friendly condition. Interestingly, the perception of right-of-way priority is crucial, as it influences how pedestrians and AVs interact and coexist in shared spaces. Some participants ($n=8$) felt that pedestrians should be prioritised in shared spaces traffic rules. They felt *PaveFlow*’s paths invaded their space, challenging their right-of-way and reducing their satisfaction. Conversely, participants ($n=6$) believed that without clear rules in shared spaces, neither pedestrians nor vehicles have absolute priority, suggesting mutual yielding as a courteous gesture. Yielding to AVs was viewed as a generous act, enhancing their satisfaction.

5 DISCUSSION

5.1 Effectiveness of *PaveFlow* in AV-Pedestrian Interaction

Multi-eHMI significantly outperformed the *Baseline* across measures, robustly supporting Hypothesis 1 (H1). This confirms that *PaveFlow*, characterised by its continuity, predictability, and flexible right-of-way, effectively enhances interactions between AVs and

pedestrians in shared spaces. These findings confirm prior research showing that eHMIs enhance pedestrian safety perception, trust, and decision-making in AV interactions [14, 19, 34]. While eHMIs are generally preferred over baseline conditions without explicit communication, our study extends these insights to shared spaces with multiple AVs, where pedestrian movement is more dynamic and right-of-way is often ambiguous.

However, qualitative findings indicated that when multiple AVs simultaneously change their motion states—such as accelerating, decelerating, stopping, or adjusting their trajectories—*PaveFlow*’s path projection proved less intuitive and offers limited benefits in reducing cognitive workload. This reveals a key challenge with dynamic feedback: while continuous path updates aid in anticipating AV movements, rapid or simultaneous changes can overwhelm pedestrians, complicating path interpretation. This observation underscores a critical design challenge for eHMIs in shared spaces with multiple AV—how to maintain clarity and usability during AV motion state transitions [44].

5.2 Scalability of PavFlow in Complex Environments

Multi-eHMI and *Single-eHMI* showed somewhat similar ratings for perceived safety, trust, and user experience, with a slight preference for *Multi-eHMI*. However, notable differences in workload were found, suggesting limitations in PavFlow's scalability in more complex environments, leading to the rejection of Hypothesis 2 (H2). This was further supported by interview findings, where participants noted that denser path projections and motion state transitions in *Multi-eHMI* complicated decision-making compared to *Single-eHMI*.

A closer examination of these challenges reveals two paradoxes. First, signals perceived as intuitive in *Single-eHMI* were seen as less intuitive in *Multi-eHMI*. The intuitiveness of eHMI diminishes as the number of eHMIs increases, highlighting the importance of evaluating the scalability of eHMI design concepts [11, 43]. Second, the flexible right-of-way principle of the design, intended to enhance AV-pedestrian interactions in complex environments, becomes problematic in *Multi-eHMI*. Real-time eHMI responses, while promoting responsiveness, also increased pedestrian workload [12]. One potential improvement is adapting the timing of PavFlow's projected paths based on the density of AVs and pedestrians. For example, in high-density environments, a slightly delayed or smoothed path projection could prevent excessive, rapid signal updates that may overwhelm pedestrians.

Finally, to put these challenges of *Multi-eHMI* into perspective and better understand their impact, we need to look at the findings on the effectiveness of *Multi-eHMI* compared to *Baseline* (H1) and *Single-eHMI* (H2) holistically, as they reveal an important nuance of scalability: while the effectiveness of eHMIs diminishes as their number increases, their presence still offers clear benefits over having no eHMI at all. This nuance meaningfully contributes to the ongoing debate over whether eHMIs are merely a gimmick or a necessity [8].

5.3 Study Limitations

The controlled VR experiment may not fully capture real-world factors such as weather, road surfaces, and unpredictable pedestrian behaviours, which may impact PavFlow's effectiveness. Technical limitations, such as processing delays and projection clarity in real-world settings, were also not considered. Regarding sample size, the culturally specific participant pool and demographic homogeneity (e.g., strong gender imbalance) may limit generalisability. Finally, while PavFlow was compared to a no-eHMI baseline, the absence of comparisons with existing eHMI designs makes it unclear whether PavFlow provides significant advantages over alternative communication methods.

6 CONCLUSION

This paper presents the design and evaluation of PavFlow, a projection-based eHMI that externalises an AV's intended path and dynamically adapts to pedestrians' movements in shared spaces. Results show that PavFlow significantly improves AV-pedestrian interactions in multi-vehicle settings compared to scenarios without it. However, as the number of AVs increases, its scalability is

challenged by the increased workload caused by motion state transitions, highlighting the need to reassess PavFlow's communication design and explore solutions to mitigate its negative impacts. This study contributes to a broader understanding of eHMI scalability and the design of effective communication strategies for AVs in shared spaces.

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References

- [1] Pavlo Bazilinskyy, Dimitra Dodou, and Joost de Winter. 2020. External Human-Machine Interfaces: Which of 729 Colors Is Best for Signaling 'Please (Do not) Cross'? In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, Toronto, ON, Canada, 3721–3728. doi:10.1109/SMC42975.2020.9282998
- [2] Fanta Camara, Patrick Dickinson, and Charles Fox. 2021. Evaluating pedestrian interaction preferences with a game theoretic autonomous vehicle in virtual reality. *Transportation research part F: traffic psychology and behaviour* 78 (2021), 410–423.
- [3] Fanta Camara and Charles Fox. 2021. Space invaders: Pedestrian proxemic utility functions and trust zones for autonomous vehicle interactions. *International Journal of Social Robotics* 13, 8 (2021), 1929–1949.
- [4] Fanta Camara and Charles Fox. 2022. Unfreezing autonomous vehicles with game theory, proxemics, and trust. *Frontiers in Computer Science* 4 (2022), 969194.
- [5] Alex Cao, Keshav K Chintamani, Abhilash K Pandya, and R Darin Ellis. 2009. NASA TLX: Software for assessing subjective mental workload. *Behavior research methods* 41 (2009), 113–117.
- [6] Luca Crosato, Chongfeng Wei, Edmond SL Ho, and Hubert PH Shum. 2021. Human-centric autonomous driving in an av-pedestrian interactive environment using svo. In *2021 IEEE 2nd International Conference on Human-Machine Systems (ICHMS)*. IEEE, Magdeburg, Germany, 1–6.
- [7] Koen De Clercq, Andre Dietrich, Juan Pablo Núñez Velasco, Joost De Winter, and Riender Happee. 2019. External human-machine interfaces on automated vehicles: Effects on pedestrian crossing decisions. *Human factors* 61, 8 (2019), 1353–1370.
- [8] Joost de Winter and Dimitra Dodou. 2022. External human-machine interfaces: Gimmick or necessity? *Transportation research interdisciplinary perspectives* 15 (2022), 100643.
- [9] Debargha Dey, Azra Habibovic, Andreas LÄfken, Philipp Wintersberger, Bastian Pflöging, Andreas Riener, Marieke Martens, and Jacques Terken. 2020. Taming the eHMI jungle: A classification taxonomy to guide, compare, and assess the design principles of automated vehicles' external human-machine interfaces. *Transportation Research Interdisciplinary Perspectives* 7 (2020), 100174. doi:10.1016/j.trip.2020.100174
- [10] Debargha Dey, Azra Habibovic, Bastian Pflöging, Marieke Martens, and Jacques Terken. 2020. Color and Animation Preferences for a Light Band eHMI in Interactions Between Automated Vehicles and Pedestrians. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376325
- [11] Debargha Dey, Arjen van Vastenhoven, Raymond H. Cuijpers, Marieke Martens, and Bastian Pflöging. 2021. Towards Scalable eHMIs: Designing for AV-VRU Communication Beyond One Pedestrian. In *13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Leeds, United Kingdom) (AutomotiveUI '21). Association for Computing Machinery, New York, NY, USA, 274–286. doi:10.1145/3409118.3475129
- [12] Yke Bauke Eisma, Lucas van Gent, and Joost de Winter. 2023. Should an external human-machine interface flash or just show text? A study with a gaze-contingent setup. *Transportation research part F: traffic psychology and behaviour* 97 (2023), 140–154.
- [13] Mitsubishi Electric. 2015. Mitsubishi electric introduces road-illuminating directional indicators.
- [14] Stefanie M Faas, Lesley-Ann Mathis, and Martin Baumann. 2020. External HMI for self-driving vehicles: Which information shall be displayed? *Transportation research part F: traffic psychology and behaviour* 68 (2020), 171–186.

- [15] Evelyn Florentine, Mark Adam Ang, Scott Drew Pendleton, Hans Andersen, and Marcelo H. Ang. 2016. Pedestrian Notification Methods in Autonomous Vehicles for Multi-Class Mobility-on-Demand Service. In *Proceedings of the Fourth International Conference on Human Agent Interaction* (Biopolis, Singapore) (HAI '16). Association for Computing Machinery, New York, NY, USA, 387–392. doi:10.1145/2974804.2974833
- [16] Xinyue Gui, Koki Toda, Stela Hanbyeol Seo, Chia-Ming Chang, and Takeo Igarashi. 2022. “I am going this way”: Gazing Eyes on Self-Driving Car Show Multiple Driving Directions. In *Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Seoul, Republic of Korea) (AutomotiveUI '22). Association for Computing Machinery, New York, NY, USA, 319–329. doi:10.1145/3543174.3545251
- [17] Azra Habibovic, Victor Malmsten Lundgren, Jonas Andersson, Maria Klingegård, Tobias Lagström, Anna Sirkka, Johan Fagerlönn, Claes Edgren, Rikard Fredriksson, Stas Krupenia, et al. 2018. Communicating intent of automated vehicles to pedestrians. *Frontiers in psychology* 9 (2018), 1336.
- [18] Marius Hoggenmueller, Martin Tomitsch, and Stewart Worrall. 2022. Designing Interactions With Shared AVs in Complex Urban Mobility Scenarios. *Frontiers in Computer Science* 4 (2022). doi:10.3389/fcomp.2022.866258
- [19] Kai Holländer, Ashley Colley, Christian Mai, Jonna Häkkinä, Florian Alt, and Bastian Pfleger. 2019. Investigating the Influence of External Car Displays on Pedestrians' Crossing Behavior in Virtual Reality. In *Proceedings of the 21st International Conference on Human-Computer Interaction with Mobile Devices and Services* (Taipei, Taiwan) (MobileHCI '19). Association for Computing Machinery, New York, NY, USA, Article 27, 11 pages. doi:10.1145/3338286.3340138
- [20] Kai Holländer, Marius Hoggenmüller, Romy Gruber, Sarah Theres Völkel, and Andreas Butz. 2022. Take It to the Curb: Scalable Communication Between Autonomous Cars and Vulnerable Road Users Through Curbstone Displays. *Frontiers in Computer Science* 4 (2022), 844245.
- [21] Liqun Huang, Runqi Chai, Kaiyuan Chen, Senchun Chai, Yuanqing Xia, and Guo-Ping Liu. 2024. Bidirectional neural network for trajectory planning: An application to medical emergency vehicle. *Neurocomputing* 591 (2024), 127763. doi:10.1016/j.neucom.2024.127763
- [22] Suresh Kumar Jayaraman, Chandler Creech, Dawn M Tilbury, X Jessie Yang, Anuj K Pradhan, Katherine M Tsui, and Lionel P Robert Jr. 2019. Pedestrian trust in automated vehicles: Role of traffic signal and AV driving behavior. *Frontiers in Robotics and AI* 6 (2019), 117.
- [23] K. Jens and J.S. Gregg. 2021. How design shapes space choice behaviors in public urban and shared indoor spaces- A review. *Sustainable Cities and Society* 65 (2021), 102592. doi:10.1016/j.scs.2020.102592
- [24] Jiun-Yin Jian, Ann M Bisantz, and Colin G Drury. 2000. Foundations for an empirically determined scale of trust in automated systems. *International journal of cognitive ergonomics* 4, 1 (2000), 53–71.
- [25] Maria Kabtoul. 2021. *Proactive and social navigation of autonomous vehicles in shared spaces*. Ph.D. Dissertation. Université Grenoble Alpes [2020-...].
- [26] Auttapone Karndacharuk, Douglas J Wilson, and Roger Dunn. 2014. A review of the evolution of shared (street) space concepts in urban environments. *Transport reviews* 34, 2 (2014), 190–220.
- [27] Harmish Khambhaita and Rachid Alami. 2020. Viewing Robot Navigation in Human Environment as a Cooperative Activity. In *Robotics Research*, Nancy M. Amato, Greg Hager, Shawna Thomas, and Miguel Torres-Torriti (Eds.). Springer International Publishing, Cham, 285–300.
- [28] Yee Mun Lee, Ruth Madigan, Chinebuli Uzundu, Jorge Garcia, Richard Romano, Gustav Markkula, and Natasha Merat. 2022. Learning to interpret novel eHMI: The effect of vehicle kinematics and eHMI familiarity on pedestrian crossing behavior. *Journal of safety research* 80 (2022), 270–280.
- [29] Mercedes-Benz. 2015. The Mercedes-Benz F 015 Luxury in Motion. <https://www.mercedes-benz.com/en/mercedes-benz/innovation/research-vehicle-f-015-luxury-in-motion/>
- [30] Trung Thanh Nguyen, Kai Holländer, Marius Hoggenmueller, Callum Parker, and Martin Tomitsch. 2019. Designing for Projection-based Communication between Autonomous Vehicles and Pedestrians. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Utrecht, Netherlands) (AutomotiveUI '19). Association for Computing Machinery, New York, NY, USA, 284–294. doi:10.1145/3342197.3344543
- [31] Manon Prédhumeau, Anne Spalanzani, and Julie Dugdale. 2021. Pedestrian behavior in shared spaces with autonomous vehicles: An integrated framework and review. *IEEE Transactions on Intelligent Vehicles* 8, 1 (2021), 438–457.
- [32] Manon Prédhumeau, Anne Spalanzani, and Julie Dugdale. 2023. Pedestrian Behavior in Shared Spaces With Autonomous Vehicles: An Integrated Framework and Review. *IEEE Transactions on Intelligent Vehicles* 8, 1 (2023), 438–457. doi:10.1109/TIV.2021.3116436
- [33] Michael Rettenmaier, Moritz Pietsch, Jonas Schmidler, and Klaus Bengler. 2019. Passing through the Bottleneck - The Potential of External Human-Machine Interfaces. In *2019 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, Paris, France, 1687–1692. doi:10.1109/IVS.2019.8814082
- [34] Alexandros Rouchitsas and Håkan Alm. 2019. External human-machine interfaces for autonomous vehicle-to-pedestrian communication: A review of empirical work. *Frontiers in psychology* 10 (2019), 2757.
- [35] Borja Ruiz-Apilánez, Kayvan Karimi, Irene García-Camacha, and Raúl Martín. 2017. Shared space streets: Design, user perception and performance. *Urban Design International* 22 (2017), 267–284.
- [36] Gabriele Scali and Robert D. Macredie. 2019. Shared Mental Models as a Way of Managing Transparency in Complex Human-Autonomy Teaming. In *Proceedings of the 14th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2019) - HUCAPP, INSTICC, SciTePress*, Prague, Czech Republic, 153–159. doi:10.5220/0007484801530159
- [37] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. 2017. Design and evaluation of a short version of the user experience questionnaire (UEQ-S). *International Journal of Interactive Multimedia and Artificial Intelligence* 4, 6 (December 2017), 103–108.
- [38] Ali Hassan Shah, Xiaodong Sun, and Yandan Lin. 2022. Study the Effect of eHMI Projection Distance and Contrast on People Acceptance in Blind-Spot Detection Scenario. *Applied Sciences* 12, 13 (2022), 6730.
- [39] Jinjuan She, Jack Neuhoof, and Qingcong Yuan. 2021. Shaping pedestrians' trust in autonomous vehicles: an effect of communication style, speed information, and adaptive strategy. *Journal of Mechanical Design* 143, 9 (2021), 091401.
- [40] Maged Shoman, Tarek Sayed, and Suliman Gargoum. 0. Transformer-Based Model for Predicting Trajectories in Autonomous Vehicle-Pedestrian Conflicts: A Proactive Approach to Road Safety. *Canadian Journal of Civil Engineering* 0, ja (0), null. doi:10.1139/cjce-2024-0137 arXiv:https://doi.org/10.1139/cjce-2024-0137
- [41] Phani Teja Singamaneni, Pilar Bachiller-Burgos, Luis J. Manso, Anaïs Garrell, Alberto Sanfeliu, Anne Spalanzani, and Rachid Alami. 2024. A survey on socially aware robot navigation: Taxonomy and future challenges. *The International Journal of Robotics Research* 43, 10 (2024), 1533–1572. doi:10.1177/02783649241230562 arXiv:https://doi.org/10.1177/02783649241230562
- [42] Tram Thi Minh Tran, Callum Parker, Marius Hoggenmüller, Yiyuan Wang, and Martin Tomitsch. 2024. Exploring the Impact of Interconnected External Interfaces in Autonomous Vehicles on Pedestrian Safety and Experience. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 89, 17 pages. doi:10.1145/3613904.3642118
- [43] Tram Thi Minh Tran, Callum Parker, and Martin Tomitsch. 2023. Scoping Out the Scalability Issues of Autonomous Vehicle-Pedestrian Interaction. In *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Ingolstadt, Germany) (AutomotiveUI '23). Association for Computing Machinery, New York, NY, USA, 167–177. doi:10.1145/3580585.3607167
- [44] Himanshu Verma, Guillaume Pythoud, Grace Eden, Denis Lalanne, and Florian Evéquo. 2019. Pedestrians and visual signs of intent: towards expressive autonomous passenger shuttles. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–31.
- [45] Yiyuan Wang, Luke Hespanhol, Stewart Worrall, and Martin Tomitsch. 2022. Pedestrian-Vehicle Interaction in Shared Space: Insights for Autonomous Vehicles. In *Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Seoul, Republic of Korea) (AutomotiveUI '22). Association for Computing Machinery, New York, NY, USA, 330–339. doi:10.1145/3543174.3546838
- [46] Annette Werner. 2018. New Colours for Autonomous Driving: An Evaluation of Chromaticities for the External Lighting Equipment of Autonomous Vehicles. *Colour Turn* 1, 1 (2018), 1–14. doi:10.25538/tct.v0i1.692
- [47] Ming-Yuan Yu, Ram Vasudevan, and Matthew Johnson-Roberson. 2020. Risk Assessment and Planning with Bidirectional Reachability for Autonomous Driving. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, Paris, France, 5363–5369. doi:10.1109/ICRA40945.2020.9197491
- [48] Xinyan Yu, Marius Hoggenmüller, and Martin Tomitsch. 2023. Your Way Or My Way: Improving Human-Robot Co-Navigation Through Robot Intent and Pedestrian Prediction Visualisations. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (Stockholm, Sweden) (HRI '23). Association for Computing Machinery, New York, NY, USA, 211–221. doi:10.1145/3568162.3576992
- [49] Xinyan Yu, Marius Hoggenmüller, Tram Thi Minh Tran, Yiyuan Wang, and Martin Tomitsch. 2024. Understanding the Interaction between Delivery Robots and Other Road and Sidewalk Users: A Study of User-generated Online Videos. *ACM Transactions on Human-Robot Interaction* 13, 4 (2024), 1–32.
- [50] Haochen Zhang, Yiyuan Wang, and Tram Thi Minh Tran. 2024. External Speech Interface: Effects of Gendered and Aged Voices on Pedestrians' Acceptance of Autonomous Vehicles in Shared Spaces. In *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Stanford, CA, USA) (AutomotiveUI '24 Adjunct). Association for Computing Machinery, New York, NY, USA, 190–196. doi:10.1145/3641308.3685046